

- Emily E Wrenbeck, Matthew A Bedewitz, Justin R Klesmith, Syeda Noshin, Cornelius S Barry, and Timothy A Whitehead. An automated Data-Driven pipeline for improving heterologous enzyme expression. *ACS Synthetic Biology*, 8(3):474–481, March 2019.
- Nicholas C. Wu, Arthur P. Young, Laith Q. Al-Mawsawi, C. Anders Olson, Jun Feng, Hangfei Qi, Shu-Hwa Chen, I-Hsuan Lu, Chung-Yen Lin, Robert G. Chin, Harding H. Luan, Nguyen Nguyen, Stanley F. Nelson, Xinmin Li, Ting-Ting Wu, and Ren Sun. High-throughput profiling of influenza A virus hemagglutinin gene at single-nucleotide resolution. *Scientific Reports*, 4(1):4942, December 2014. ISSN 2045-2322. doi: 10.1038/srep04942. URL <https://www.nature.com/articles/srep04942>.
- Nicholas C. Wu, C. Anders Olson, Yushen Du, Shuai Le, Kevin Tran, Roland Remenyi, Danyang Gong, Laith Q. Al-Mawsawi, Hangfei Qi, Ting-Ting Wu, and Ren Sun. Functional Constraint Profiling of a Viral Protein Reveals Discordance of Evolutionary Conservation and Functionality. *PLOS Genetics*, 11(7):e1005310, July 2015. ISSN 1553-7404. doi: 10.1371/journal.pgen.1005310. URL <https://dx.plos.org/10.1371/journal.pgen.1005310>.
- Nicholas C Wu, Lei Dai, C Anders Olson, James O Lloyd-Smith, and Ren Sun. Adaptation in protein fitness landscapes is facilitated by indirect paths. *eLife*, 5:e16965, July 2016. ISSN 2050-084X. doi: 10.7554/eLife.16965. URL <https://elifesciences.org/articles/16965>.
- Yingzhou Wu, Hanqing Liu, Roujia Li, Song Sun, Jochen Weile, and Frederick P Roth. Improved pathogenicity prediction for rare human missense variants. *The American Journal of Human Genetics*, 108(10):1891–1906, 2021.
- Zachary Wu, S. B. Jennifer Kan, Russell D. Lewis, Bruce J. Wittmann, and Frances H. Arnold. Machine learning-assisted directed protein evolution with combinatorial libraries. *Proceedings of the National Academy of Sciences*, 116:8852 – 8858, 2019. URL <https://api.semanticscholar.org/CorpusID:67770057>.
- Michael J Xie, Gareth A Cromie, Katherine Owens, Martin S Timour, Michelle Tang, J Nathan Kutz, Ayman W El-Hattab, Richard N McLaughlin, and Aimée M Dudley. Predicting the functional effect of compound heterozygous genotypes from large scale variant effect maps. *bioRxiv*, January 2023.
- Minghao Xu, Zuobai Zhang, Jiarui Lu, Zhaocheng Zhu, Yangtian Zhang, Chang Ma, Runcheng Liu, and Jian Tang. PEER: A Comprehensive and Multi-Task Benchmark for Protein Sequence Understanding, September 2022. URL <http://arxiv.org/abs/2206.02096>. arXiv:2206.02096 [cs].
- Kevin Kaichuang Yang, Zachary Wu, and Frances H. Arnold. Machine-learning-guided directed evolution for protein engineering. *Nature Methods*, pages 1–8, 2018. URL <https://api.semanticscholar.org/CorpusID:128342395>.
- Kevin Kaichuang Yang, Alex X. Lu, and Nicolás Fusi. Convolutions are competitive with transformers for protein sequence pretraining. *bioRxiv*, 2023a. URL <https://api.semanticscholar.org/CorpusID:248990392>.
- Kevin Kaichuang Yang, Niccolò Zanichelli, and Hugh Yeh. Masked inverse folding with sequence transfer for protein representation learning. *bioRxiv*, 2023b. URL <https://api.semanticscholar.org/CorpusID:249241961>.
- Sook Wah Yee, Christian Macdonald, Darko Mitrovic, Xujia Zhou, Megan L Koleske, Jia Yang, Dina Buitrago Silva, Patrick Rockefeller Grimes, Donovan Trinidad, Swati S More, Linda Kachuri, John S Witte, Lucie Delemotte, Kathleen M Giacomini, and Willow Coyote-Maestas. The full spectrum of OCT1 (SLC22A1) mutations bridges transporter biophysics to drug pharmacogenomics. *bioRxiv*, June 2023.
- Heather J. Young, Matthew Chan, Balaji Selvam, Steven K. Szymanski, Diwakar Shukla, and Erik Procko. Deep Mutagenesis of a Transporter for Uptake of a Non-Native Substrate Identifies Conformationally Dynamic Regions. preprint, Biochemistry, April 2021. URL <http://biorxiv.org/lookup/doi/10.1101/2021.04.19.440442>.
- Haicang Zhang, Michelle S Xu, Xiao Fan, Wendy K Chung, and Yufeng Shen. Predicting functional effect of missense variants using graph attention neural networks. *Nature Machine Intelligence*, 4(11):1017–1028, 2022.
- Naihui Zhou, Yuxiang Jiang, Timothy Bergquist, Alexandra J. Lee, Balint Z. Kacsoh, Alex Crocker, Kimberley A. Lewis, George E. Georgiou, Huy N. Nguyen, Nafiz Imtiaz Bin Hamid, Larry Davis, Tunca Dogan, Volkan Atalay, Ahmet Sureyya Rifaoglu, Alperen Dalkiran, Rengul Cetin-Atalay, Chengxin Zhang, Rebecca L. Hurto, Peter L. Freddolino, Yang Zhang, Prajwal Bhat, Fran Supek, José María Fernández, Branislava Gemović, Vladimir Perovic, Radoslav Davidovic, Neven Sumonja, Nevena Veljkovic, Ehsaneddin Asgari, Mohammad R. K. Mofrad, Giuseppe Profiti, Castrense Savojardo, Pier Luigi Martelli, Rita Casadio, Florian

Boecker, Indika Kahanda, Natalie Thurlby, Alice Mchardy, Alexandre Renaux, Rabie Saidi, Julian Gough, Alex Alves Freitas, Magdalena Antczak, Fábio Fabris, Mark N. Wass, Jie Hou, Jianlin Cheng, Zheng Wang, Alfonso E. Romero, Alberto Paccanaro, Haixuan Yang, Tatyana Goldberg, Chenguang Zhao, Liisa Holm, Petri Törönen, Alan Medlar, Elaine Zosa, Itamar Borukhov, Ilya B. Novikov, Angela D. Wilkins, Olivier Lichtarge, Po-Han Chi, Wei-Cheng Tseng, Michal Linial, Peter W. Rose, Christophe Dessimoz, Vedrana Vidulin, Sašo Džeroski, Ian P. W. Sillitoe, Sayoni Das, Jonathan G. Lees, David T. Jones, Cen Wan, Domenico Cozzetto, Rui Fa, Mateo Torres, Alex Warwick Vesztrocy, Jose Manuel Rodriguez, Michael L. Tress, Marco Frasca, Marco Notaro, Giuliano Grossi, Alessandro Petrini, Matteo Ré, Giorgio Valentini, Marco Mesiti, Daniel B. Roche, Jonas Reeb, David W. Ritchie, Sabeur Aridhi, Seyed Ziaeddin Alborzi, Marie-Dominique Devignes, Da Chen Emily Koo, Richard Bonneau, Vladimir Gligorijević, Meet Barot, Hai Fang, Stefano Toppo, Enrico Lavezzo, Marco Falda, Michele Berselli, Silvio C. E. Tosatto, Marco Carraro, Damiano Piovesan, Hafeez ur Rehman, Qizhong Mao, Shanshan Zhang, Slobodan Vucetic, Gage S Black, Dane Jo, Dallas J. Larsen, Ashton Omdahl, Luke Sagers, Erica Suh, Jonathan B. Dayton, Liam James McGuffin, Danielle Allison Brackenridge, Patricia C. Babbitt, Jeffrey M. Yunes, Paolo Fontana, Feng Zhang, Shanfeng Zhu, Ronghui You, Zihan Zhang, Suyang Dai, Shuwei Yao, Weidong Tian, Renzhi Cao, Caleb Chandler, Miguel Amezola, Devon Johnson, Jia-Ming Chang, Wen-Hung Liao, Yi-Wei Liu, Stefano Pascarelli, Yotam Frank, R. Hoehndorf, Maxat Kulmanov, Imane Boudellioua, Gianfranco Politano, Stefano Di Carlo, Alfredo Benso, Kai Hakala, Filip Ginter, Farokh Mehryary, Suwisa Kaewphan, Jari Björne, Hans Moen, Martti Tolvanen, Tapio Salakoski, Daisuke Kihara, Aashish Jain, Tomislav Šmuc, Adrian M. Altenhoff, Asa Ben-Hur, Burkhard Rost, Steven E. Brenner, Christine A. Orengo, Constance J. Jeffery, Giovanni Bosco, Deborah A. Hogan, Maria Jesus Martin, Claire O'Donovan, Sean D. Mooney, Casey S. Greene, Predrag Radivojac, and Iddo Friedberg. The CAFA challenge reports improved protein function prediction and new functional annotations for hundreds of genes through experimental screens. *Genome Biology*, 20, 2019.

Julia Zinkus-Boltz, Craig DeValk, and Bryan C. Dickinson. A Phage-Assisted Continuous Selection Approach for Deep Mutational Scanning of Protein–Protein Interactions. *ACS Chemical Biology*, 14(12):2757–2767, December 2019. ISSN 1554-8929, 1554-8937. doi: 10.1021/acscchembio.9b00669. URL <https://pubs.acs.org/doi/10.1021/acscchembio.9b00669>.

A Appendix

A.1 Social Impact

Protein design holds considerable promise for various fields, ranging from medicine to agriculture, and is likely to have a profound social impact. However, the development of such technology introduces several concerns, particularly relating to the dual use of protein fitness and design models. For instance, while beneficial for areas like drug design, these models can also be potentially utilized for harmful purposes such as bio-weapon design. Consider a generative model developed for therapeutic purposes: it typically penalizes predicted toxicity. Yet, the logic of this model could be inverted to instead reward for toxicity [Urbina et al., 2022]. Indeed, any tool or benchmark developed to improve protein design can be manipulated for nefarious objectives. Lastly, protein fitness models will significantly influence the way experiments are conducted. With increased adoption and development of protein design, substantial portions of experimental work can be accelerated, leading to quicker iterations and improved results. Nonetheless, the need for wet lab experimentation remains. These technological advancements will serve to augment, rather than completely supplant, traditional experimental procedures.

Additionally, the American College of Medical Genetics (ACMG) disregards computational prediction of variant effects due to insufficient validation. Consequently, it is essential to create benchmarks using clinical data in order to promote the acceptance of these machine learning methods in medical practice.

A.2 Limitations

Deep mutational scans While significant efforts have been dedicated to curating and preprocessing a diverse set of deep mutational scans (DMS), the very nature of these scans imposes biases and limitations to this benchmark:

1. **Measurement noise** Experiments do not have a perfect dynamic range, often imposing a restrictive ceiling and/or floor to the measured response of mutation effects that is not meaningful for protein function and mutation effect prediction. Furthermore, noise is a perennial issue in high-throughput assays, and some assays have poor experimental replicate correlation. Taken together, this means that one cannot expect perfect correlation between experiment and model. Since these considerations affect different proteins to different extents, computing average Spearman correlations across proteins can be misleading.
2. **Bias** There is additional bias in the types of proteins chosen for deep mutational scans. This can be due both to experimental limitations on which proteins’ functions can be assayed (for example, disordered proteins are challenging), and to protein prioritisation considerations (for example, viral and cancer-related proteins are over-represented).
3. **Representativeness** No assay is fully representative of the impacts of protein changes on the evolutionary fitness of an organism, which typically involves a convolution of molecular functions across changing environments. In fact, many assays target only a single feature such as expression, binding, or enzymatic activity.
4. **Inconsistent processing** The reported fitness effects from DMS are themselves the result of modeling and analysis of the raw data. The treatment of data is extremely heterogeneous across the community and different analyses can lead to different conclusions on the effect of mutations. For a perfect standardised curation of experimental results one should treat all data with the same approach. We leave this type of analysis for future work.

Human mutation databases ClinVar data has the advantage of covering more proteins than DMS, even if only human proteins involved in disease. But it has several limitations:

1. **Noise** This dataset, by the very community-based nature of it, is very noisy. Filtering to more stringently curated ClinVar labels, or to more recent labels, improves correlation with predictions from sequence models [Frazer et al., 2021]. Here we decided to keep a reasonable number of clinical labels – a trade-off between quantity and quality.
2. **Bias** Clinical labels are biased towards classes of proteins that are heavily studied, such as well-known cancer genes, as well as towards European ancestry.
3. **Circularity** Grimm et al. [2015] details two types of circularity that hinder the evaluation of human variant effect predictors. In a supervised benchmark, there is the potential for data leakage from training to testing, even for different variants in the same protein. Even for our unsupervised benchmark, where models have not trained on clinical labels, there is the potential for another type of circularity: evolutionary conservation is one of the criteria used to classify a variant as benign or pathogenic in ClinVar.

Finally, the current benchmarks are limited to mutations in coding regions. But there are both DMS datasets and clinical labels (although fewer of them) in regulatory regions – for example in UTRs, introns, promoters. This could be an interesting direction of growth for these benchmarks.

A.3 Datasets

A.3.1 DMS assays

Evolution of protein fitness benchmarks based on DMS assays As discussed in § 2, our DMS benchmarks build on several prior works that had compiled a growing library of such assays. We summarize their content in Table A1.

Category	Mut. Type	Metric or Setting	EVmutation	Deep Sequence	ProteinGym v0.1	ProteinGym v1.0
DMS	Sub.	Assays (mut.)	26 (0.1M)	38 (0.7M)	87 (1.6M)	217 (2.4M)
	Ind.	Assays (mut.)	0 (0k)	0 (0k)	7 (270k)	66 (289k)
Clinical	Sub.	Genes (mut.)	0	0	0	2,525 (63k)
	Ind.	Genes (mut.)	0	0	0	1,555 (3k)
Training regime	Sub.	Zero-shot	✓	✓	✓	✓
	Ind.	Zero-shot	-	-	✓	✓
	Sub.	Supervised	-	-	-	✓
	Ind.	Supervised	-	-	-	✓
Baselines	Sub.	Zero-shot	5	3	9	42
	Ind.	Zero-shot	0	0	3	20
	Sub.	Supervised	0	0	0	9
	Ind.	Supervised	0	0	0	3
Metrics	-	Zero-shot	2	3	3	5
	-	Supervised	0	0	0	2

Table A1: **Evolution of protein fitness benchmarks** ProteinGym v0.1 corresponds to benchmarks in Notin et al. [2022a], while ProteinGym v1.0 corresponds to benchmarks in this paper. The EVmutation benchmark was introduced in Hopf et al. [2017], while the DeepSequence benchmark was developed in Riesselman et al. [2018]. Sub., Ind. and mut. are shorthands for substitutions, indels and mutants respectively.

Selection and processing We focused on several different criteria when determining which DMS assays to include in ProteinGym. These are:

1. The public availability of data
2. The experimental throughput (how many mutations were assayed)
3. The level of noise between experiment replicates
4. The dynamic range of the assay
5. The assay type (selection, enrichment, etc) and whether or not it captures evolutionary constraints.
6. If the assay used amino-acid substitution or indel mutations (no UTR, tRNA, promoter, etc. variants were included).

Final list of assays In-depth metadata about the assays, including the assay type, UniProt ID, MSA start and end positions, mutated positions, and target sequence, is provided under the `reference_files` directory in the codebase. A complete list of included assays is presented at the end of the appendix (See Tables A19 and A20)

Processing of large thermostability dataset A large dataset of thermostability assays of 331 natural domains [Tsuboyama et al., 2023] contributed 65 assays to our list. We processed these assays as follows:

We used the set of non-redundant natural domains (referred to as Dataset #5 in the original paper). After mapping to UniProt IDs for our DMS id naming convention and removing datasets where none of the tested evolutionary models had a Spearman correlation above 0.2 (suggesting that there is inadequate evolutionary fitness signal in the stability assay, preventing meaningful comparisons between models), we were left with 65 thermostability

Function type	# Assays		Description
	Subs	Indels	
Activity	43	3	Assays that directly or indirectly measure a protein’s catalytic (or otherwise biochemical) activity
Binding	14	0	Assays that measure the affinity or the degree to which a protein binds its target
Expression	17	2	Assays that measure how much the protein is expressed in a cell
Organismal fitness	77	6	Assays that measure how much changes in the protein affect an organism’s growth rate
Stability	66	55	Assays that measure how thermostable a protein is

Table A2: **DMS assay function types.** The number of substitution and indel assays in each of the 5 function type categories and a general description used to categorize the assays.

assays of short domains (40-72 residues long). For substitutions, there was 99+% coverage of each position with 14-19 mutations per position (and 52 of those datasets with multiples), and for indels there was a deletion, Gly and Ala insertion at every position.

Classification of DMS assays We grouped the substitution DMS assays into five function types: activity, binding, expression, organismal fitness, and stability, assigning each to a primary class such that the classes are non-overlapping. We provide a brief description of each class in Table A2. We took into account multiple factors to delineate the groups as cleanly as possible, most importantly the type of experiment used (e.g., cell growth, cell sorting, biochemical, stability, etc.). Some assay types presented ambiguities that were resolved as follows:

- **Cell growth** Growth-based assays link the function of their target protein to cellular survival and growth. These assays generally fall under “Organismal fitness”, particularly for complementation assays where mutants are tested for their ability to replace the natural function of the wild-type target and allow cell growth. However, in some cases, cell growth is artificially linked to readouts such as enzymatic activity (e.g., SRC_HUMAN [Ahler et al., 2019], MET_HUMAN [Estevam et al., 2023], Q837P4_ENTFA [Meier et al., 2023]), expression (HXXK4_HUMAN [Gersing et al., 2023]), or binding (RASK_HUMAN [Weng et al., 2022]). We recategorized assays that were deemed sufficiently distinct from the protein’s natural function in the cell.
- **Cell sorting** Cell sorting, most commonly fluorescence-activated cell sorting (FACS), artificially selects cells that have been fluorescently labeled according to their function. This class of experiments generally falls into “Activity”, “Binding”, or “Expression”, depending on the choice of labeling target. “Expression” assays label the target protein itself, quantifying its expression levels inside the cell or on the cell surface (e.g., NUD15_HUMAN [Suiter et al., 2020], OPSD_HUMAN [Wan et al., 2019]). Since protein abundance and cell surface presentation are dependent on protein stability, these assays are often characterized elsewhere as stability assays (e.g., PRKN_HUMAN [Clausen et al., 2023], PTEN_HUMAN [Matreyek et al., 2021]). “Activity” assays quantify levels of an enzyme’s product or reporters of enzymatic activity (e.g., OXDA_RHOTO [Vanella et al., 2023], A0A247D711_LISMN [Stadelmann et al., 2021], PPARG_HUMAN [UK Monogenic Diabetes Consortium et al., 2016]). “Binding” assays quantify levels of a binding partner or another reporter of protein binding (e.g., ACE2_HUMAN [Chan et al., 2020], GCN4_YEAST [Staller et al., 2018]).

Cross-validation schemes As described in § 4.2, we leverage the 3 types of cross-validation schemes (Random, Contiguous and Modulo) introduced in Notin et al. [2023] for the different analyses in the supervised regime. For the Random split, we randomly assigned each mutant to one of 5 folds. The Contiguous scheme is obtained by splitting the sequence in contiguous segments along its length, ensuring the segments are comprised of the same number of positions. We only consider positions mutated, which may not span the entire length of the protein sequence. The Modulo scheme is obtained by assigning positions to each fold using the modulo of the position number by the total number of folds. Therefore, for a 5-fold cross-validation, position 1 is assigned to fold 1, position 2 to fold 2, ..., position 6 to fold 1, etc. Once again, we make sure to only consider mutated positions. We operate a five fold cross-validation for all assays except for assays F7YBW8_MESOW [Aakre et al., 2015] and SPG1_STRSG [Wu et al., 2016], as these contain only 4 mutated positions. Note that multiple mutants generally involve several positions that may not be easily separated into the independent folds as discussed in the Contiguous and Modulo schemes above. Similarly, indels do not lend themselves well to these two cross-validation schemes. Thus, we only keep single mutants for all supervised analyses related to substitutions, and only focus on the Random cross-validation scheme for all indels analyses.